

$f(z) = Sz$ be zero. Applying Euler's method to the same system results in

$$z_{n+1} = z_n + hSz_n = (I + hS)z_n,$$

and the condition for Euler's method to conserve volume is that $\det(I + hS) = 1$. The conditions for volume preservation by the flow map and its Euler approximation are essentially unrelated. Thus Euler's method does not in general conserve phase space volume (it conserves volume only in very special cases—see Exercise 6).

Yet it turns out that there are some numerical methods that always conserve the volume when this is conserved by the flow of the differential equations, or for some particular classes of differential equations. For example, consider the planar system

$$\dot{u} = f(u, v),$$

$$\dot{v} = g(u, v),$$

satisfying the condition $\frac{\partial f}{\partial u} + \frac{\partial g}{\partial v} = 0$, and the asymmetrical variant of Euler's method defined by

$$u_{n+1} = u_n + hf(u_{n+1}, v_n),$$

$$v_{n+1} = v_n + hg(u_{n+1}, v_n).$$

The components of the Jacobian matrix can be obtained using implicit differentiation, they are:

$$a_{11} = \frac{\partial u_{n+1}}{\partial u_n}, \quad a_{12} = \frac{\partial u_{n+1}}{\partial v_n}, \quad a_{21} = \frac{\partial v_{n+1}}{\partial u_n}, \quad a_{22} = \frac{\partial v_{n+1}}{\partial v_n}.$$

So

$$a_{11} = 1 + hf_u a_{11}, \quad a_{12} = hf_u a_{12} + hf_v, \quad a_{21} = hg_u a_{11}, \quad a_{22} = 1 + hg_u a_{12} + hg_v.$$

Solving for the various entries we have

$$\mathcal{G}'_h = \begin{bmatrix} 1/(1 - hf_u) & hf_v/(1 - hf_u) \\ hg_u/(1 - hf_u) & 1 + hg_v + h^2 g_u f_v/(1 - hf_u) \end{bmatrix},$$

and calculating the determinant of the Jacobian results in

$$\det \mathcal{G}'_h = \frac{1 + hg_v}{1 - hf_u}.$$

In the event that the vector field is divergence free, we have $f_u + g_v = 0$ which implies that the numerator and denominator are identical, and it follows that

$\det \mathcal{G}'_h = 1$. Thus the asymmetric variant of the Euler method is area preserving, even though the standard Euler method is not.

It would be valuable to have a general method for deriving volume conserving methods. It turns out that volume conservation is, itself, most readily obtained as a consequence of a more fundamental property of Hamiltonian flows, the conservation of the symplectic form.

2.3.3 The Symplectic Property

Let $m = 2N_c$. A symplectic map $\Phi : \mathbb{R}^m \rightarrow \mathbb{R}^m$ is one that preserves the symplectic differential 2-form. The simplest way to write this is as the following algebraic condition on the Jacobian matrix of Φ :

$$\Phi'^T J \Phi' = J.$$

Recall that the Jacobian matrix is the $m \times m$ matrix of partial derivatives of the components of Φ , $(\Phi')_{ij} = \frac{\partial \Phi_i}{\partial z_j}$, while

$$J = \begin{bmatrix} \mathbf{0} & I \\ -I & \mathbf{0} \end{bmatrix}.$$

So the equation $\Phi'^T J \Phi' = J$ represents a large number ($4N_c^2$) of equations.

To explain the origin of this formula, we need a brief diversion into the world of differential forms. This language will also be useful in Chap. 4, where we consider constrained systems.

A 1-form α defined on $\mathbb{R}^m$ is a family of linear mappings from $\mathbb{R}^m$ to $\mathbb{R}$, defined for each point of $\mathbb{R}^m$. Let $\mathbf{a} : \mathbb{R}^m \rightarrow \mathbb{R}^m$, then we may define a one-form associated to this vector by $\alpha(\mathbf{x})(\xi) = \mathbf{a}(\mathbf{x})^T \xi$.

The differential of a function $g : \mathbb{R}^m \rightarrow \mathbb{R}$, denoted dg , is a family of linear mappings (one for each point in phase space) from vectors $\xi \in \mathbb{R}^m$ into the reals defined by

$$dg(\mathbf{q}, \mathbf{p})(\xi) = \nabla g(\mathbf{q}, \mathbf{p})^T \xi.$$

So, denoting the i th position coordinate by q_i , we have $dq_i(\xi) = \xi_i$; the differential is thus an example of a 1-form.

On the other hand the so called *wedge product* of 1-forms α , β , is an example of a 2-form, at any point in phase space it can be viewed as a quadratic form, i.e. a scalar valued function of two vectors which is linear in each argument. It is written $\alpha \wedge \beta$ and is defined, for vectors $\xi, \eta \in \mathbb{R}^m$ by

$$\alpha \wedge \beta(\xi, \eta) = \alpha(\xi)\beta(\eta) - \alpha(\eta)\beta(\xi).$$